SOFTWARE ENGINEERING LAB – 1

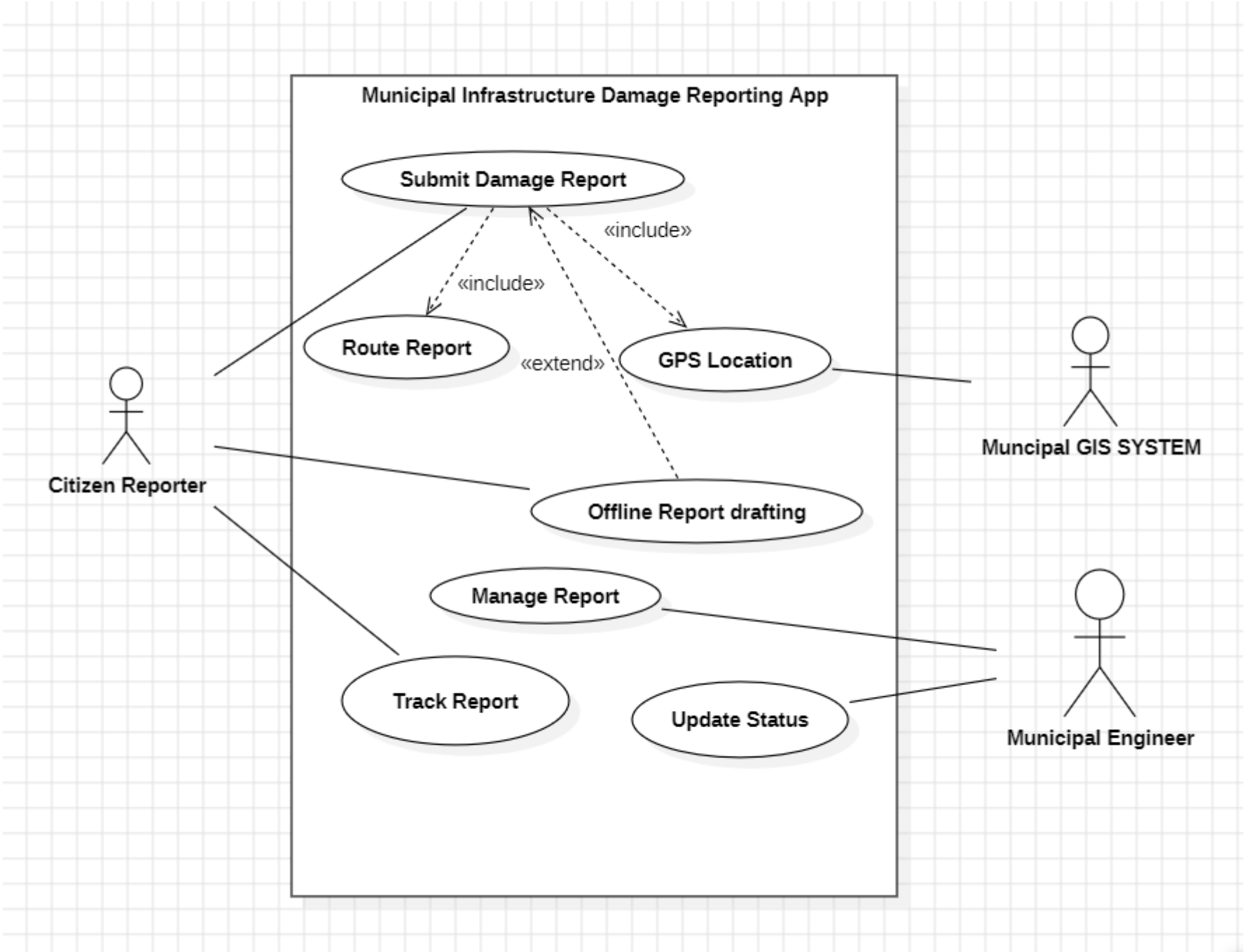
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Requirements Traceability Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall extract EXIF GPS metadata from uploaded damage photos to automatically pinpoint incident location on the municipal GIS map.	**High**	**Pass:** Pothole report is routed to the Road Infrastructure division with exact GPS coordinates. **Fail:** Report is submitted without location data.	Enables accurate incident location identification and automatic routing.
FR-002	Functional	The system shall allow a Citizen Reporter to submit a damage report by selecting the damage type, uploading a photo, and providing a description.	**High**	**Pass:** A report containing damage type, photo, and description is successfully submitted and assigned a unique report ID. **Fail:** Submission is rejected when required information is missing.	Provides the core functionality for reporting municipal infrastructure damage.
FR-003	Functional	The system shall automatically route each submitted damage report to the appropriate municipal engineering department based on the damage type.	**High**	**Pass:** A pothole report is routed to the Road Infrastructure division and a broken streetlight report is routed to the Electrical division. **Fail:** Report is routed to an incorrect department.	Ensures that each incident reaches the responsible department.
FR-004	Functional	The system shall allow a Municipal Engineer to view assigned damage reports and update their status.	**High**	**Pass:** The engineer can change a report status from "Assigned" to "In Progress" or "Resolved," and the updated status is displayed. **Fail:** The status remains unchanged after a valid update.	Enables municipal engineers to manage and track reported infrastructure issues.
FR-005	Functional	The system shall allow a Citizen Reporter to view the current status of a submitted damage report using its report ID.	**Medium**	**Pass:** Entering a valid report ID displays the report status and assigned department. **Fail:** An invalid report ID does not display another report's information.	Allows citizens to track the progress of their submitted reports.

NFR-001	Nonfunctional	The reporting portal shall support offline report drafting and auto-sync when mobile network connectivity resumes.	**High**	**Pass:** A report drafted while offline is automatically synchronized within 30 seconds after network connectivity is restored. **Fail:** The report is not synchronized after connectivity resumes.	Allows citizens to prepare reports even when network connectivity is temporarily unavailable.
NFR-002	Nonfunctional	The system shall restrict modification of damage reports to the Citizen Reporter who created the report and authorized Municipal Engineers.	**High**	**Pass:** An unauthorized user attempting to modify a report receives an access-denied response. **Fail:** An unauthorized user can modify report information.	Protects municipal reports from unauthorized modification.

UML Use-Case Diagram :


Municipal Infrastructure Damage Reporting App
Submit Damage Report
«include»
«include»
Route Report
«extend»
GPS Location
Citizen Reporter
Muncipal GIS SYSTEM
Offline Report drafting
Manage Report
Track Report
Update Status
Municipal Engineer

Use-Case Flow Specification :

Use Case : Submit Damage Report

Primary Actor: Citizen Reporter
Secondary Actors: Municipal GIS System, Municipal Engineer
Includes: Extract GPS Location, Route Damage Report
Extends: Offline Report Drafting (triggered when network connectivity is unavailable)

Preconditions

- The Citizen Reporter has access to the Municipal Infrastructure Damage Reporting App.
- The Citizen Reporter has a photograph of the infrastructure damage.
- The application is ready to accept a new damage report.

Postconditions
Success:

- A unique report ID is generated and the damage report is stored in the system.
- The incident location is recorded using GPS information.
- The report is automatically routed to the appropriate municipal engineering department.

Failure:

- The report is not submitted and the Citizen Reporter is informed of the missing or invalid information.

Main Success Scenario

1. The Citizen Reporter opens the Municipal Infrastructure Damage Reporting App.
2. The Citizen Reporter selects **Submit Damage Report**.
3. The system asks the Citizen Reporter to select the type of infrastructure damage.
4. The Citizen Reporter selects the damage type, such as pothole, broken streetlight, or water pipeline leak.
5. The Citizen Reporter uploads a photograph of the damage.

6. The system invokes **UC-02 (Extract GPS Location)** to extract EXIF GPS metadata from the uploaded photograph.
7. The system identifies the incident location using the Municipal GIS System.
8. The Citizen Reporter provides a description of the damage.
9. The Citizen Reporter submits the completed report.
10. The system validates the submitted information.
11. The system generates a unique report ID.
12. The system invokes **UC-03 (Route Damage Report)** to determine the appropriate municipal engineering department.
13. The system stores the report and displays the report ID and submission confirmation.
14. The use case ends successfully.

Alternate Flow
3a. GPS Location Unavailable

- The system cannot extract valid GPS information from the uploaded photograph.
- The system informs the Citizen Reporter that the location could not be automatically determined.
- The Citizen Reporter provides the incident location manually.
- The system validates and records the manually provided location.
- The use case continues from Step 8 of the Main Success Scenario.

Alternate Flow
5a. Network Unavailable

- The system detects that network connectivity is unavailable while the Citizen Reporter is preparing the report.
- The system activates **UC-04 (Offline Report Drafting)**.
- The Citizen Reporter continues preparing the report offline.
- When network connectivity resumes, the system automatically synchronizes the drafted report.
- The use case continues with the report validation and submission process.